



OPEN Enhancement of airflow uniformity in multi-tray dryers through a CFD-integrated sequential linear programming framework

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Drying is a widely employed preservation method in food processing, aimed at extending shelf life by reducing moisture content, thereby minimizing storage and transportation costs. Despite their widespread use, cabinet-type multi-tray dryers often exhibit non-uniform airflow distribution, leading to inconsistent airflow uniformity. This study presents an optimization framework to enhance airflow uniformity by strategically repositioning supply fans, integrating Computational Fluid Dynamics (CFD) simulations with Sequential Linear Programming (SLP). An L_{25} orthogonal array guided the CFD experiment design, and a second-order polynomial regression model was developed to correlate the airflow uniformity index (U_i) with fan positions (P_1, P_2, P_3). The optimization, constrained by fixed geometric and operational parameters including a 78.80 mm supply gap, 45° baffle angle, 2.74 m/s Inlet airflow speed and 315 mm fan diameter was conducted for a 32-tray dryer (dimensions: 1420 × 650 × 1600 mm). The regression model achieved $R^2 = 0.9247$. Incorporating trust-region techniques into the SLP framework significantly improved convergence efficiency: static and dynamic trust-region approaches reduced iterations to 15 and 8, respectively, compared to 661 without trust-region control. These methods adaptively managed step sizes, promoting faster and more stable convergence without compromising accuracy. The MATLAB-based optimization yielded optimal fan positions of $P_1 = 359.05$ mm, $P_2 = 395$ mm, and $P_3 = 395$ mm, achieving a maximum uniformity of 76.5% with 0.2% prediction error. This CFD–SLP integrated approach offers a reliable, scalable, and cost-effective solution for enhancing airflow uniformity in industrial dryers and holds promise for broader applications such as thermal or fluid process optimization tasks across food, pharmaceutical, and chemical industries.

Keywords Multi-tray cabinet dryer, Airflow uniformity, Computational fluid dynamics, Sequential linear programming, Trust-region methods, Fan positioning optimization

There is an increasing trend in the global food market industries to improve the food products value chains through different techniques¹. Drying is one of the most common food preservation techniques². Drying can minimize the storage and transportation costs, and makes handling easier by reducing size and weight. It is the process of moisture removal due to the simultaneous heat and mass transfer under controlled conditions.

Efforts have been made recently to develop dried foods with high quality by using innovative technologies. The drying technologies can reduce the water activity to a level with no microbiological activity and biochemical reactions. Most commonly, the efficiency of a dryer is dependent on air flow speed and drying air temperature^{3,4}. Based on a most recent comprehensive review⁵, Increasing the drying temperature and airflow speed significantly accelerates moisture evaporation and reduces drying time, via enhancing both heat and mass transfer.

Advancement on sun drying leads to solar drying. The artificial method of drying technologies such as cabinet, tunnel, conveyor, fluidized bed, drum and vacuum shelf. During drying, water from the surface evaporates by mass and heat transfer mechanisms resulting in a reduction of the water activity of the product^{5–9}. In a cabinet type multi tray dryer, more products can be loaded as the trays are arranged at different levels¹⁰. The key to a quality operation of those dryers is uniformity of hot airflow distribution over trays. However, multi-tray dryers (large size industrial dryers), encounter poor airflow speed uniformity. This study focused on optimizing

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the positions of the supply fans to enhance airflow distribution, while other design elements such as airflow-guiding baffles and the supply zone gap beneath the wire mesh trays were kept fixed^{5,11}.

This research specifically focused on optimizing the position of the supply fans under a fixed supply zone gap (78.80 mm) and baffle angles (45°, downward)^{12–14}. As illustrated in Fig. 2, three supply fans were positioned centrally on each side along the height of the dryer. A fan with a diameter of Ø315 mm and six impellers, each set at a 10° impeller angle (AXC 315-6/12°-4-PV axial fan, <https://www.systemair.com>), was selected based on the calculated static pressure and volume flow rate requirement (15.56 pa and 0.252 m³/s). The size and capacity of the fans and the drying unit are presented in Table 1.

CFD has been utilized in different types of food drying applications, to substitute expensive experimental trials and predict distribution of airflow speed and temperature uniformity in the drying chamber^{15–17}. CFD simulation to obtain a uniform distribution of drying air flow speed and temperature by changing the angle of baffles and position of exhaust, with three trays under electric resistance heat source shows purely uniform distribution of temperature. However, the airflow distribution ranges from 0.016 m/s to 3.89 m/s over the entire drying unit¹⁵. The most recent design and simulation of airflow speed on multi tray dryer to improve uniformity by using CFD with 18 trays shows a range of distribution of airflow speed from 0 m/s to 6 m/s over the drying unit with 6.17 m/s inlet airflow speed¹⁸. The same wise research on prediction of drying uniformity of multi tray dryer with 21 trays, were assumed as porous media with 10% porosity has simulated on CFD. Results show better uniformity of distribution of heat on 18 trays. However, the airflow speed over the drying unit varies from 0 m/s to 3 m/s with 3 m/s inlet airflow speed¹⁰.

Sequential Linear Programming (SLP), also known as Successive Linear Programming, is an iterative method used to solve nonlinear constrained optimization problems by converting them into a sequence of linear programming problems¹⁹. This approach is based on successive first-order linearizations of the objective and constraint functions. SLP has been successfully applied to optimize complex nonlinear problems in various fields, including heat distribution in living spaces²⁰, optimal selection of auxiliary fans²¹, hybrid renewable energy systems²², and pick-and-place task optimization²³, resulting in significant airflow uniformity improvements.

The study aims to strategically reposition the dryer’s supply fans to achieve more consistent airflow and enhance airflow uniformity, using CFD simulation and SLP optimization techniques. Specifically, the research focuses on:

- Developing a CFD-based fluid model of the dryer and perform mesh-independence tests for reliable airflow predictions.
- Generating surface plots and fitting regression models linking airflow uniformity index (U_i) to fan position parameters using DOE data.
- Applying SLP for optimizing fan positions and to enhance convergence using static and dynamic trust-region strategies.
- Validating the optimized solutions by comparing SLP-predicted U_i values with CFD-simulated results and quantifying accuracy using percentage error.

A second-order polynomial model, representing fan positional configurations, was derived from Computational Fluid Dynamics (CFD) results obtained via experiments based on an L_{25} orthogonal array design. The model’s behavior was analysed using nonlinear surface plots and curve fitting. Optimization of the fan positions then followed, employing SLP with static and dynamic trust-region strategies, further guided by insights from literatures on drying chamber design modifications^{7,11,15,24}.

Methodology

This study employed a structured methodology to optimize multi-tray dryer performance. The workflow (Fig. 1) outlines the process, integrating experimental design, CFD modeling with mesh verification, polynomial surface fitting, sequential linear programming-based optimization, and validation of the optimized configuration.

Experiment design (DOE)

After identifying the controlled and fixed factors (summarized in Table 4), an L_{25} (5^3) orthogonal array experimental design was implemented using Minitab statistical software. The factors P_1 , P_2 , and P_3 , corresponding to the gap between supply fans, were varied at five levels each (207, 254, 301, 348, and 395 mm), as illustrated in Fig. 2a. The airflow speed uniformity index (U_i) was selected as the dependent variable to evaluate airflow uniformity. Details of the controlled and fixed factors are provided in Table 2. For each DOE run, simulations were performed using ANSYS 2022 R1, ANSYS Fluent, and the airflow speed uniformity was quantified using the area-weighted uniformity index.

Component	Detail
Drying unit	1420 mm×650 mm×1600 mm 32 Number of trays (600×600) on two chambers i.e., 16 trays on each chamber
Supply fan	Ø315 mm, and six number of impellers with 10° impeller angle 13 pa static pressure, 0.252 m ³ /s volume flow rate with 2.5 m/s outlet airflow speed

Table 1. Specifications of the drying unit and built-in fans.

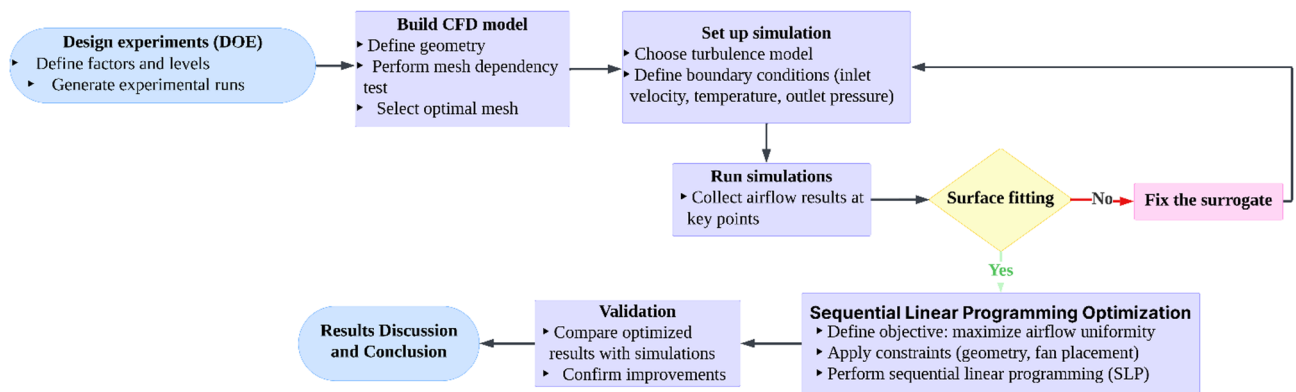


Fig. 1. Work flow to optimize multi-tray dryer performance.

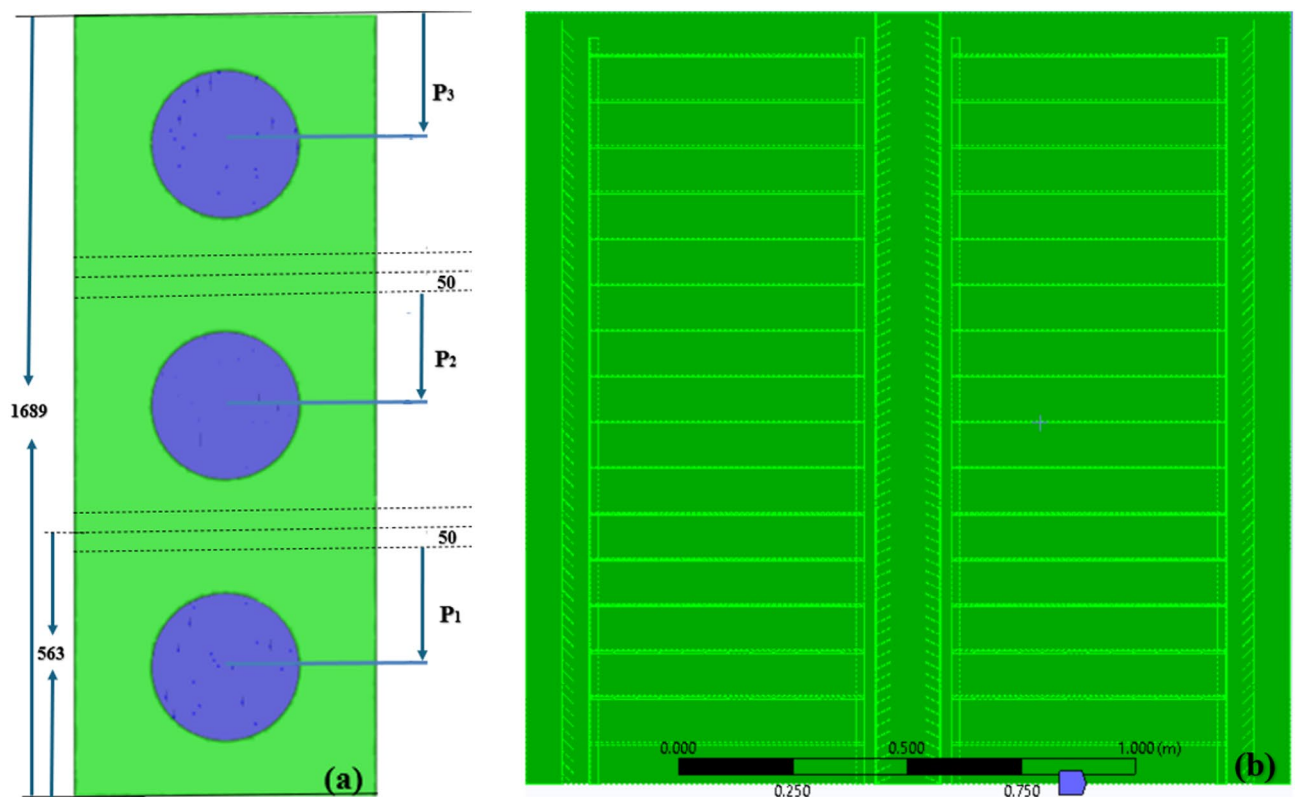


Fig. 2. (a) Position of supply fans (b) Subtracted 2D fluid model.

Controlled factors	Symbol	Factor levels (mm)	Fixed factors	Constant values
Gap between supply fans (mm)	P_1, P_2, P_3	207–395	Supply zone gap (mm)	78.80
			Inlet airflow speed (m/s)	2.74
			Baffle angle (°)	45
			Fan diameter (mm)	315

Table 2. Controlled and constant factors in the experimental design.

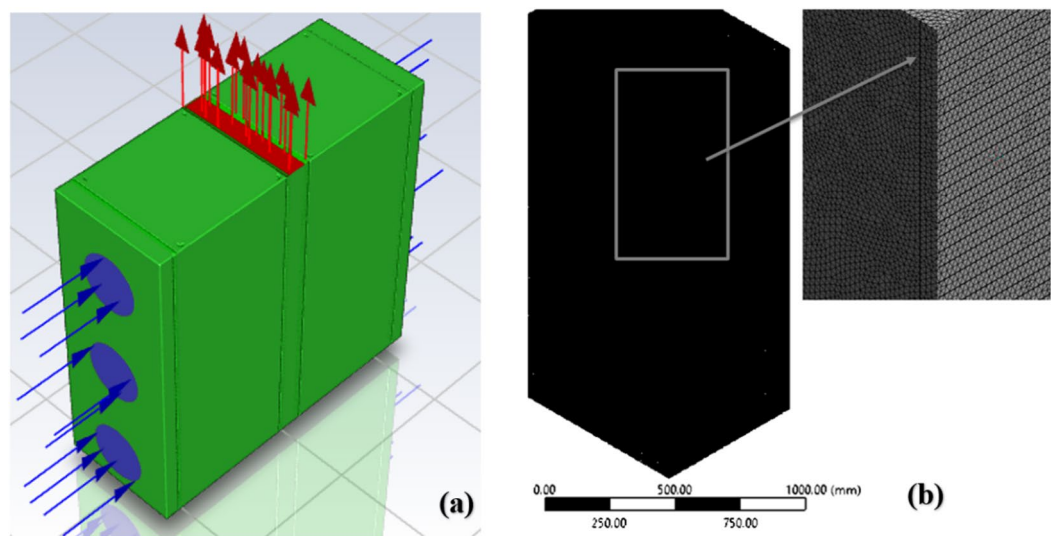


Fig. 3. (a) 3D Fluid model and boundary conditions (b) Meshed symmetry of dryer mesh size (0.01 m).

Trial	Mesh size (m)	Number of elements	Number of nodes	Air flow speed (m/s)
1	0.0125	4,190,669	883,902	2.949
2	0.010	7,801,943	1,567,607	2.719
3	0.0075	10,244,438	2,154,810	2.708

Table 3. Mesh dependency/independency test.

Fluid modelling and mesh dependency test

For each experiment design a fluid model was created by subtracting the solid structures of the dryer, such as the body, baffles, wire mesh trays, and trolley from a 2D model (Fig. 2b). The resulting geometry was saved in .igs format and imported into Design Modeler in ANSYS Fluent for each simulation run. Figure 3a illustrates the 3D fluid model of the dryer.

As shown on Table 3a mesh dependency analysis was conducted by varying mesh sizes while keeping all boundary conditions constant. Three mesh sizes 0.0125 m, 0.01 m, and 0.0075 m were tested, and airflow speed convergence was evaluated at a fixed point (L, W, H) = (1.2, 0.3, 0.6 m) on tray 8. The variation in predicted airflow speed between the two finest meshes was only 0.41%, confirming mesh independence. Consequently, the medium mesh (0.010 m, approximately 7.8 million elements) was selected as the optimal configuration, offering a balance between computational cost and solution accuracy. Figure 3 (b) shows the symmetry of dryer meshed with 0.01 m, with a zoom in sectional view.

ANSYS Fluent reported a high overall mesh quality of 0.999, while global mesh skewness metrics indicated good numerical quality, with a maximum of 0.17, average of 0.08, and minimum of 0.04, all well within the acceptable range (< 0.95), ensuring accurate and stable numerical performance. In addition, mesh adaptation was applied during the simulation to maintain mesh independence and further enhance accuracy¹⁰.

Convergence of the flow solution was verified through the residual histories shown in Fig. 4. The residuals for momentum, turbulence kinetic energy (k), and specific dissipation rate (ω) decreased below 10⁻⁴, while the continuity residual dropped by more than two orders of magnitude. These results confirm that the final solution achieved both numerical stability and physical consistency within the first 150 iterations. Based on this, all runs were conducted with a maximum of 150 iterations.

Model selection and boundary conditions setup

The inlet airflow speed was set to 2.74 m/s based on prior research, with a turbulence intensity of 5% and a turbulence length scale of 0.07 × the inlet diameter (315 mm). The outlet was specified as a pressure outlet at 0 MPa (gauge). All walls were assumed smooth, with negligible roughness effects. Baffles were represented as solid internal surfaces, with angles fixed according to previous studies. Since this study focuses on airflow rather than heat distribution, the energy equations were not considered. Air was treated as an incompressible working fluid with default thermophysical properties in ANSYS Fluent. Buoyancy effects were neglected because the flow was dominated by forced convection, and the maximum velocity corresponded to a Mach number far below 0.3.

To achieve accurate airflow predictions, selecting an appropriate turbulence model is essential. As this study does not focus on heat distribution, the energy equations were not considered. The standard k-ε model performs well for turbulent shear flows but is less accurate for flows with adverse pressure gradients, rotation, or streamline curvature. In the drying process, the presence of baffles (air diffusers), heating coils, and wire

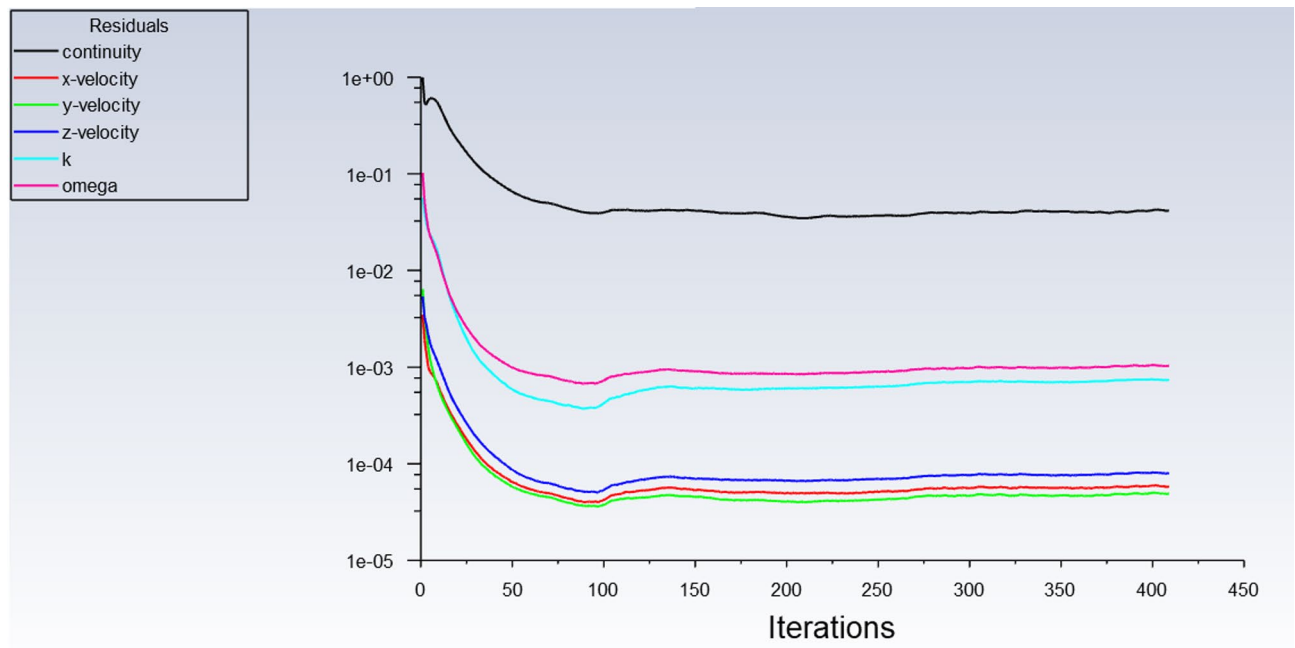


Fig. 4. Residual convergence at 0.01 m mesh.

mesh trays generates strain flows and static pressure differences, where the k- ϵ model may not predict airflow circulation accurately. The k- ω model is more effective for boundary layers with adverse pressure gradients and separated flows, and it can also handle turbulent flows in porous media^{10,25}. An improved version, the SST k- ω model, combines the advantages of k- ϵ and k- ω models across the domain. Therefore, the SST k- ω model was selected to accurately predict airflow over the walls and wire mesh trays.

Surface plotting and fitting

Based on the U_i results corresponding to each DOE point, surface plots of U_i versus P_1 , P_2 and P_3 were generated with origin 2018 software to gain insight into the input-output relationships. A 4D surface plot was constructed. Subsequently, polynomial nonlinear surface fitting was performed, and a second-order polynomial regression model with interaction terms for P_1 , P_2 and P_3 was developed using the analysis of variance (ANOVA) technique.

Sequential linear programming-based optimization and validation

This study employs a Sequential Linear Programming (SLP) in MATLAB R2024b to optimize the positions of supply fans in a multi-tray cabinet dryer, with the objective of maximizing the airflow uniformity index (U_i). The optimization framework was implemented in MATLAB, beginning with an initial guess for the design variables (P_1 , P_2 , P_3). The design variables were iteratively updated based on a small step size ($\alpha=0.01$), which ensured stability and reduced the risk of overshooting the optimum solution. Convergence was determined when the norm of the design variable update between successive iterations fell below a tolerance threshold ($\text{tol}=1\text{e-}6$), with a maximum of 1000 iterations allowed to facilitate sufficient exploration of the design space.

In addition to this baseline SLP configuration, an enhanced SLP framework incorporating trust-region strategies was established to further regulate the step size and directionality of the search process. This method constrains the updates within a trust region defined by a radius vector Δ , ensuring that steps remain within a bounded local neighbourhood. The optimization incorporated an exponentially decaying step size mechanism, where the step size α was reduced over iterations according to a decay rate ($\alpha_0 = 1$, decay rate = 0.95), with a lower bound ($\alpha_{\min} = 0.01$) to prevent stagnation. This adaptive strategy enables large, exploratory steps during the early iterations and progressively finer updates as convergence nears, facilitating both rapid progress and precise refinement. Two distinct trust-region schemes were implemented to guide the optimization process. In the Static Trust Region approach, the trust region radius decays progressively over iterations. A step is accepted if the improvement ratio (ρ) defined as the ratio of actual to predicted improvement is positive ($\rho > 0$). If the step is rejected ($\rho \leq 0$), the trust region radius is reduced by 5%, with a lower bound set at 0.01 to maintain sufficient search flexibility. In contrast, the Dynamic Trust Region method adaptively adjusts the radius based on model prediction accuracy. When $\rho > 0.75$, indicating a high degree of agreement between predicted and actual improvements, the trust region is expanded to allow broader exploration. Conversely, if $\rho < 0.25$, suggesting poor model fidelity, the region is contracted to enforce more conservative steps. This adaptive mechanism allows the optimizer to respond to the local landscape of the objective function, promoting stability and ensuring tighter convergence near the optimum.

Importantly, the decaying step size and trust region mechanisms operate independently, with the former enforcing global control over the step magnitude, and the latter enforcing local directional constraints.

At each iteration, the nonlinear objective function (U_i) is locally approximated using a linear model. The gradient of the objective function is computed using central finite differences, and the resulting linear programming subproblem is solved using MATLAB's *linprog* function. The solver minimizes the negative gradient while respecting the bounds on the design variables and the trust region constraints.

Finally, to evaluate the fidelity of the SLP model, predicted U_i values were compared against the actual values computed via CFD simulations at the optimal fan positions by using the percentage error metric defined in Eq. (1). Additionally, a parity plot comparing the predicted and CFD-obtained airflow speed uniformity index (U_i) confirms the reliability and practical applicability of the proposed SLP-based optimization framework for enhancing airflow uniformity in drying systems.

$$\text{Percentage Error} = \frac{U_i \text{ predicted} - U_i \text{ actual}}{U_i \text{ predicted}} \times 100\% \tag{1}$$

Results and discussion
Polynomial nonlinear surface fitting

Based on the CFD results, the positions of the supply fans at 25 design points designated as P_1, P_2, P_3 were evaluated for their effect on the airflow speed uniformity index (U_i). The corresponding values of P_1, P_2, P_3 , and U_i are presented in Table 4.

As shown in Fig. 5b, 4D surface plotting was performed using the design of experiments (DOE) data before fitting the model to visually analyse the relationship between the input parameters (P_1, P_2, P_3) and the response variable (U_i). This visualization provided insight into how changes in factor levels influence the output nonlinearly, supporting the selection of a suitable regression model. As shown in Fig. 5a, each coloured surface illustrates how the uniformity index (U_i) varies with P_2 and P_3 while holding P_1 constant at one of five selected design values. Comparing these surfaces reveals that the overall landscape of U_i shifts upward as P_1 increases, with the lowest values occurring around $P_1 = 207$ mm, and peak uniformity emerging near $P_1 = 348$ mm.

Based on the data presented in Table 4 and the surface plot in Fig. 5 (b), a second-order polynomial regression model with interaction terms for P_1, P_2 , and P_3 was developed. The resulting fitted surface, shown alongside the actual data points (Fig. 6), illustrates the accuracy of the model in capturing the variation in the uniformity index (U_i). The mathematical expression of this relationship is provided in Eq. (2). The model achieved a coefficient of determination R^2 of 0.9247, indicating that approximately 92.47% of the variability with a high level of predictive reliability (95%).

Run	P_1	P_2	P_3	U_i actual	U_i Predicted
1	207	207	207	0.7160	0.7160
2	207	254	254	0.7141	0.7141
3	207	301	301	0.7247	0.7247
4	207	348	348	0.7159	0.7159
5	207	395	395	0.7325	0.7325
6	254	207	254	0.7391	0.7391
7	254	254	301	0.7368	0.7368
8	254	301	348	0.7425	0.7425
9	254	348	395	0.7449	0.7449
10	254	395	207	0.7090	0.7090
11	301	207	301	0.7323	0.7323
12	301	254	348	0.7445	0.7445
13	301	301	395	0.7492	0.7492
14	301	348	207	0.7234	0.7234
15	301	395	254	0.7303	0.7303
16	348	207	348	0.7368	0.7368
17	348	254	395	0.7489	0.7489
18	348	301	207	0.7328	0.7328
19	348	348	254	0.7475	0.7475
20	348	395	301	0.7414	0.7414
21	395	207	395	0.7293	0.7293
22	395	254	207	0.7299	0.7299
23	395	301	254	0.7405	0.7405
24	395	348	301	0.7418	0.7418
25	395	395	348	0.7545	0.7545

Table 4. Airflow speed uniformity results corresponding to each experiment design.

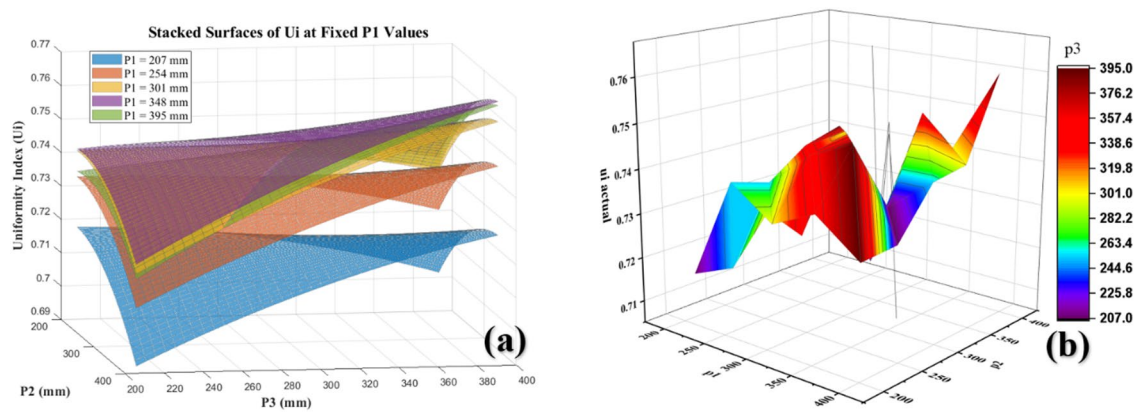


Fig. 5. Position of supply fans Vs Ui surface plots (a) stacked surfaces plot of U_i at fixed P_1 values (b) 4D surface plot.

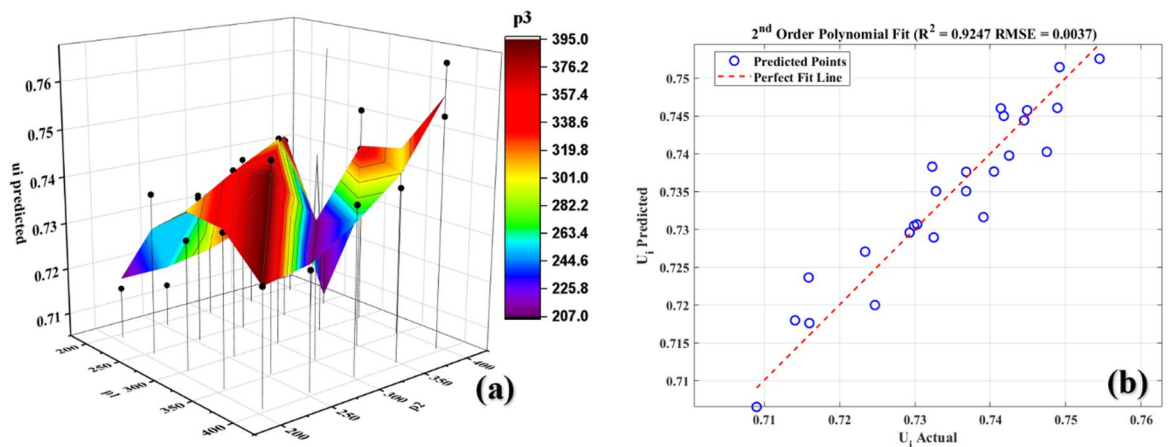


Fig. 6. (a) 4D surface fitting with a Polynomial regression model (b) Model's accuracy.

$$U_i = 0.6230 + 0.0008995P_1 - 0.00009913P_2 - 0.0002526P_3 + 3.364 \times 10^{-7}P_1P_2 + 4.926 \times 10^{-7}P_1P_3 + 1.609 \times 10^{-7}P_2P_3 - 1.645 \times 10^{-6}P_1^2 - 7.249 \times 10^{-7}P_2^2 - 4.269 \times 10^{-7}P_3^2 - 3.183 \times 10^{-10}P_1P_2P_3 \quad (2)$$

Figure 6b shows the comparison between predicted and actual values of the airflow speed uniformity index (U_i). The scatter plot demonstrates how closely the model's predictions align with the observed CFD results. The red dashed line representing an ideal fit (Predicted U_i =Actual U_i), where perfect prediction would lie. The points cluster closely around this line, indicating good model accuracy and consistency, further supported by a high R^2 value (0.9247) and RMSE of 0.0037. The corresponding data used for this plot is presented in Table 4.

Optimization

Sequential linear programming was employed to determine the optimum positions of the supply fans, considering the airflow uniformity index (U_i) as the objective function and P_1 , P_2 and P_3 as design constraints.

$$\begin{aligned} \text{Maximize } U_i &= 0.6230 + 0.0008995P_1 - 0.00009913P_2 - 0.0002526P_3 \\ &+ 3.364 \times 10^{-7}P_1P_2 + 4.926 \times 10^{-7}P_1P_3 + 1.609 \times 10^{-7}P_2P_3 \\ &- 1.645 \times 10^{-6}P_1^2 - 7.249 \times 10^{-7}P_2^2 - 4.269 \times 10^{-7}P_3^2 - 3.183 \times 10^{-10}P_1P_2P_3 \end{aligned} \quad (3)$$

$$\text{Such that : } \left. \begin{aligned} 207 &\leq P_1 \leq 395 \\ 207 &\leq P_2 \leq 395 \\ 207 &\leq P_3 \leq 395 \end{aligned} \right\} \quad (4)$$

As a result of using SLP with and without trust region strategies, the optimal positions of the supply fans in the multi-tray cabinet dryer were successfully identified. By considering a strict convergence tolerance ($\text{tol} = 1e^{-6}$), the nonlinear objective function (U_i) was efficiently maximized. Compared to the SLP method without trust

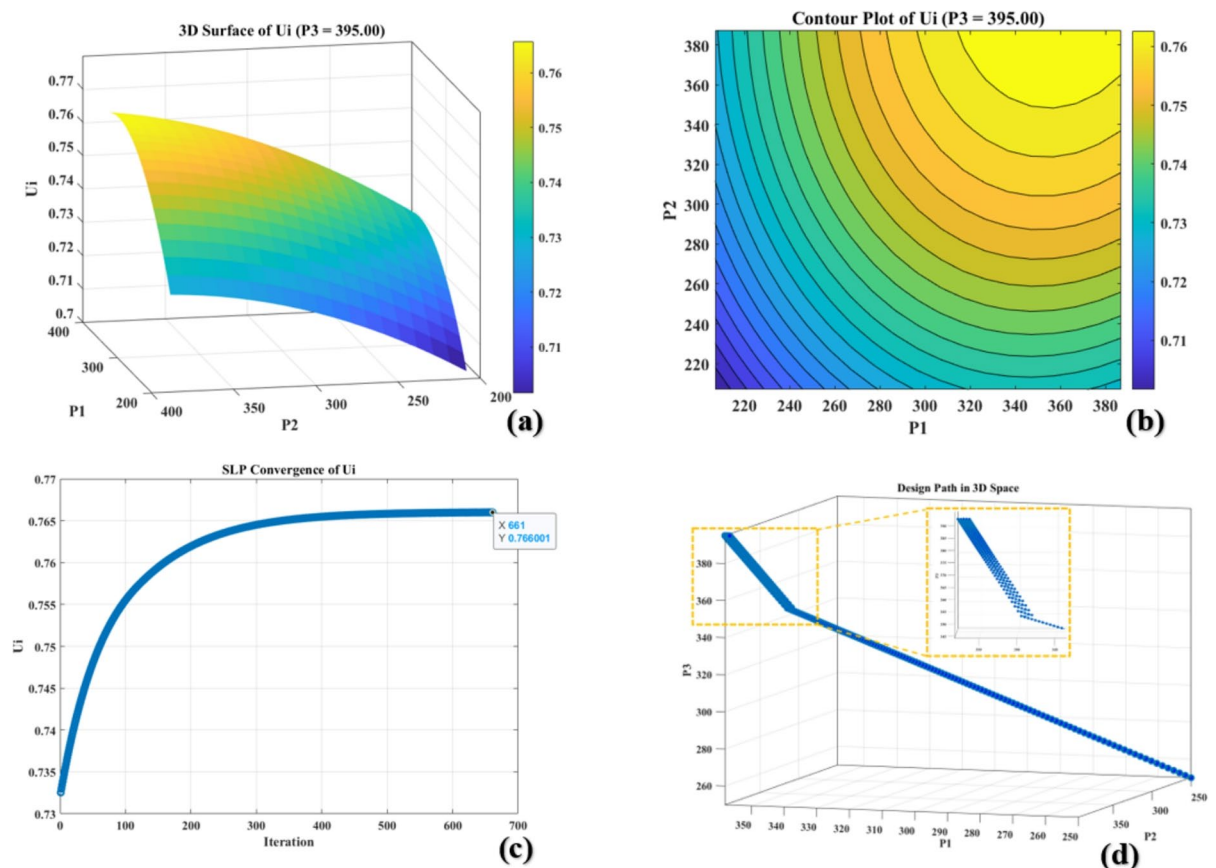


Fig. 7. (a) Static trust region based SLP optimization surface plot. (b) Static trust region based SLP optimization contour plot. (c) Baseline SLP (SLP without trust region) Convergence of U_i . (d) Baseline SLP (SLP without trust region) design points. (e) Static trust region based SLP Convergence of U_i . (f) Static trust region based SLP design path. (g) Dynamic trust region based SLP Convergence of U_i . (h) Dynamic trust region based SLP design path.

region, which required approximately 650 iterations to converge (Fig. 7c), SLP with static trust region approach together with adaptive step size (decreasing in every iteration) decayed the region size at each iteration and prevents getting stuck in small updates, Fig. 7e. While, the dynamic approach has adaptively expanded or contracted the trust region based on the step acceptance criteria. As illustrated in Fig. 7g, This approach ensured convergence in fewer iterations and provided a straightforward design path. Both trust-region methods demonstrated substantially faster convergence compared to the standard SLP approach without trust-region control, which required approximately 661 iterations to converge (Fig. 7c), following a complex zigzagging path (Fig. 7d). In contrast, the trust-region strategies facilitated a more stable and directed optimization process with minimal iterations. A steady increase in the U_i value was observed, characterized by rapid improvements during early iterations and gradual stabilization as the algorithm approached the optimum. This behavior is attributed to the use of a small step size, which played a crucial role in regulating search direction, minimizing the risk of overshooting, and preventing oscillations around local optima. The carefully chosen parameters ensured an effective balance between convergence speed and solution quality, enabling the algorithm to closely approximate the global maximum. Among trust region-based methods, the lowest prediction error (0.201%) was achieved, indicating high accuracy of the optimization outcome within the defined design bounds.

In contrast, the SLP with a static trust region converged in just 15 iterations (Fig. 7e), and the dynamic trust region approach achieved convergence in only 8 iterations (Fig. 7g). As revealed on Table 5, these trust region strategies did not significantly alter the optimal solution values. As shown in Fig. 7g and h, the dynamic trust region method provided the fastest convergence and the smoothest optimization path, exhibiting minimal zigzagging behavior compared to the static trust region strategy (Fig. 7f). Overall, incorporating a trust region particularly a dynamic one substantially improved convergence speed and efficiency, as shown in Table 5.

The SLP algorithm with dynamic trust region ultimately converged to an optimal configuration with fan positions of $P_1 = 359.0482$ mm, $P_2 = 395$ mm, and $P_3 = 395$ mm, achieving a maximum U_i value of 0.76604. This result demonstrates both the stability and effectiveness of SLP in navigating and optimizing within a nonlinear design space. When compared to recent studies^{26–28}, a significant improvement in airflow uniformity has been achieved.

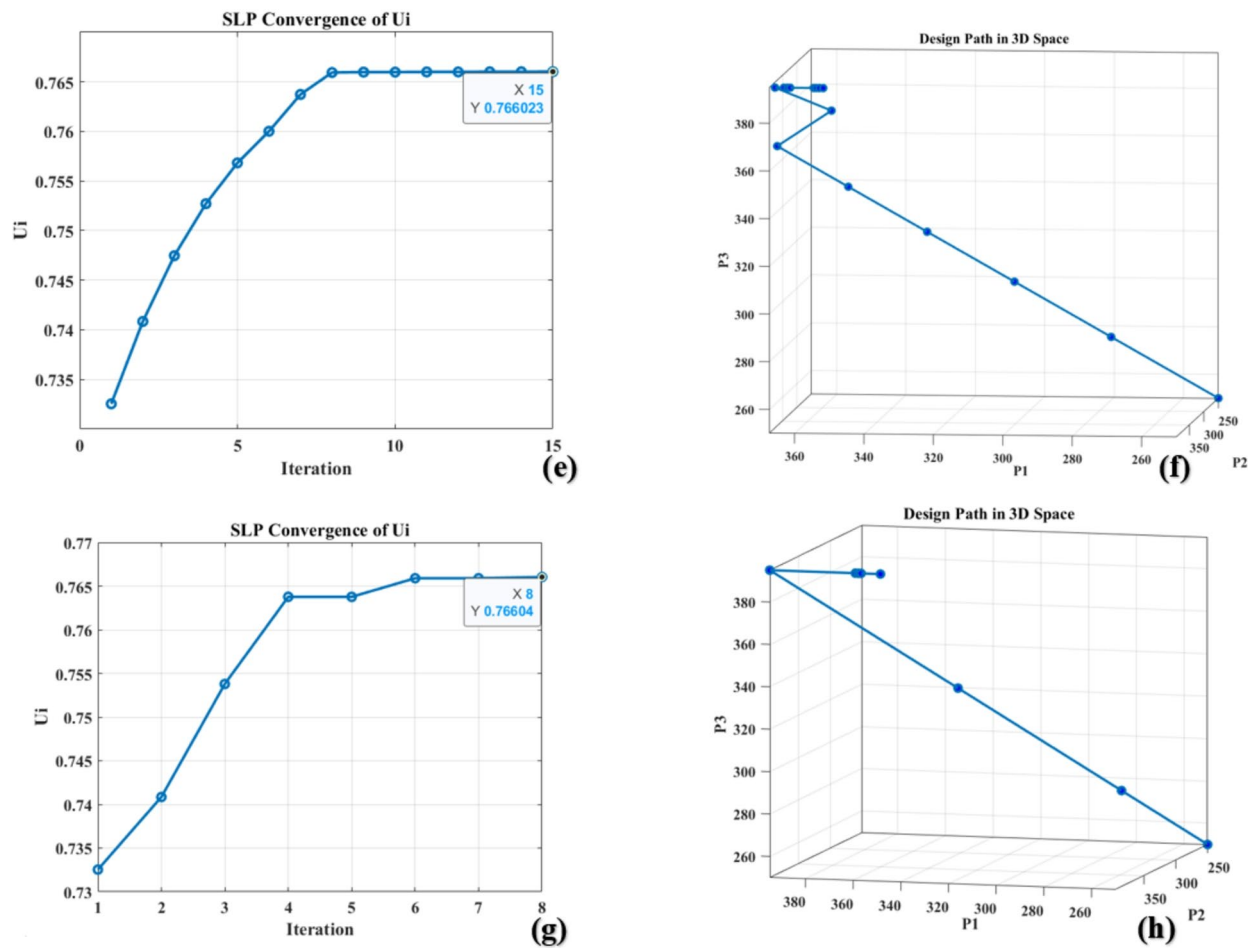


Fig. 7. (continued)

Methods	P_1	P_2	P_3	U_i Predicted	U_i Actual	% Error	CPU time (seconds)
SLP without trust region	357.7562	394.8724	394.8724	0.76601	0.7645	0.201	1.0692
SLP with Static trust region	354.4648	395.0000	395.0000	0.76602	0.7633	0.356	0.0858
SLP with Dynamic trust region	359.0482	395.0000	395.0000	0.76604	0.7642	0.241	0.0757

Table 5. Performance comparison of SLP optimization methods.

Further insights were gained through 3D surface and contour plots (Fig. 7a and b) respectively, through fixing P_3 at its optimal value and varying P_1 and P_2 . These plots clearly depict a smooth, unimodal response surface, with the U_i value increasing toward a peak in the region around the optimized coordinates. The smooth curvature and absence of abrupt changes suggest that the linear approximations used in SLP are valid within the local neighbourhoods explored at each iteration. This confirms the effectiveness of SLP in handling moderately nonlinear systems and highlights its suitability for real-world engineering optimization problems, where computational efficiency and interpretability are important.

Validation

To validate the model’s predictive capability beyond the optimized region, additional DOE points were evaluated by randomly selecting P_1 , P_2 , and P_3 within the range of 207–395 mm. Figure 8 presents a parity plot comparing the predicted and CFD-obtained airflow speed uniformity index (U_i), at 95% confidence intervals. The scatter points closely follow the red dashed line representing the ideal fit, indicating strong agreement between the surrogate model and CFD results. This confirms good generalization performance, with $R^2 = 0.9315$ and $RMSE = 0.0021$.

Although the original objective function (U_i) is nonlinear and derived from CFD-based approximations, the Sequential Linear Programming (SLP) method, which employs linear approximations using gradient information at each iteration successfully approximated the optimal solution. As shown in Fig. 8, the parity plot demonstrates close agreement between the predicted and actual U_i values, while Table 5 indicates a very small prediction

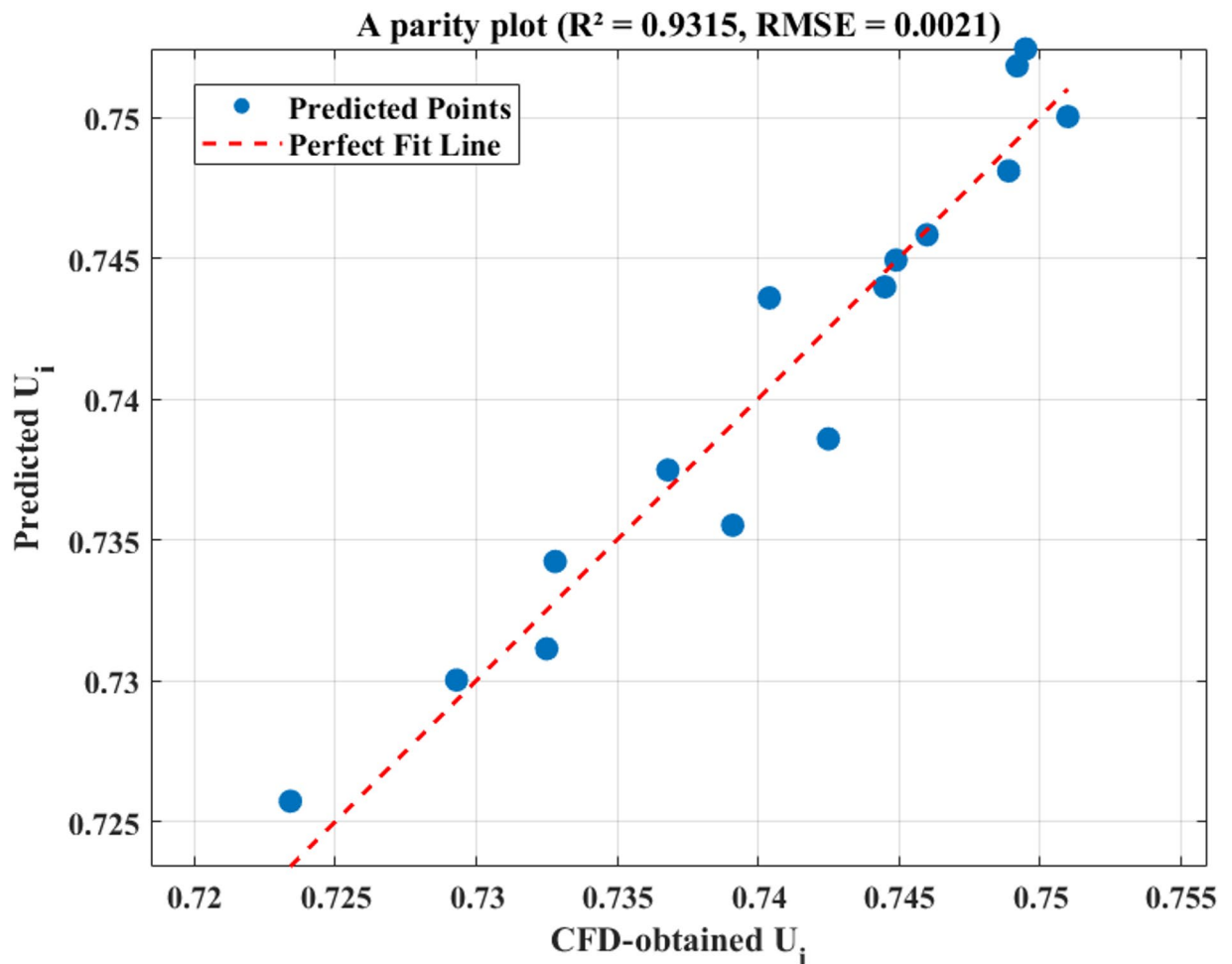


Fig. 8. Parity plot comparing predicted and CFD-obtained airflow speed uniformity index (U_i).

error. The SLP optimization achieved near-optimal supply fan positions (P_1 , P_2 , and P_3) without overshooting or instability. Notably, the SLP approach without trust-region strategies yielded a minimum error of 0.201% and required 1.069 s of CPU time. Introducing a static trust region reduced the error to 0.356% with a CPU time of 0.0858 s, while the dynamic trust region approach achieved an error of 0.241% in only 0.0757 s. These results demonstrate that trust-region strategies, particularly dynamic adjustment, not only maintain accuracy but also substantially reduce computational cost. The consistently low errors across all approaches further confirm the reliability of the regression-based U_i function employed in the optimization process.

As shown in Fig. 9 (a), high-velocity regions remain concentrated near the outlet walls, while the central zones experience reduced speeds due to flow dissipation and recirculation. The upper and lower trays are more exposed to velocity variations, with localized zones of both high-speed jets and low-speed recirculation, which diminish overall uniformity. In contrast, moving farther from the center of the supply fans along the z-axis (Fig. 9 (b)) results in a more balanced distribution, with airflow across the 32 trays (16 on each side) becoming relatively uniform. This configuration produces a consistent velocity range of 0.82–1.9 m/s, providing favourable drying conditions and marking a significant improvement over recent studies^{4,26–29}.

Nevertheless, residual non-uniformities persist. Low-speed recirculation near the lower trays can lead to under-drying, while high-speed jet impingement near the upper trays may cause localized over-drying. These effects highlight the practical limitations of the current optimization: even with careful fan placement, achieving full uniformity is constrained by the inherent flow physics, including boundary layer effects, momentum dissipation, and geometric asymmetries within the cabinet^{12,14,30}.

To further mitigate these residual non-uniformities, future geometric modifications could be explored beyond fan positioning. Potential strategies including fine-tuning baffle angles and configurations to guide airflow more evenly, introducing flow straighteners or perforated panels to reduce jet impingement, and adjusting tray spacing or inlet/outlet shapes to minimize recirculation zones. Implementing such modifications has been shown to improve both airflow speed and temperature uniformity^{7,15,16,31}, bringing multi-tray cabinet dryers closer to ideal drying conditions. Table 6 presents a comparative summary of recent related studies, highlighting similarities in the optimization of supply fan positions and other geometrical parameters for enhanced airflow distribution in multi-tray cabinet dryers.

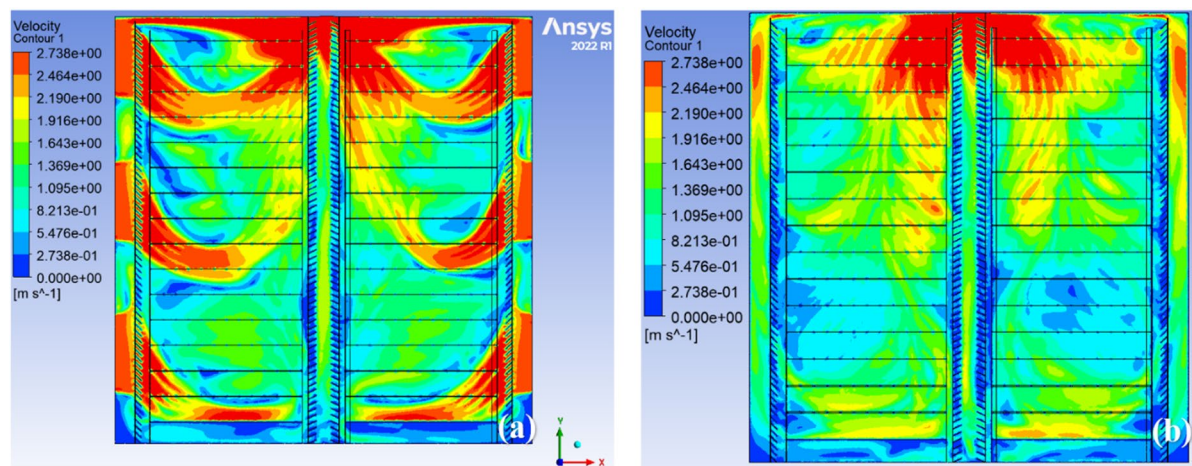


Fig. 9. Airflow velocity contour plots along the z-axis: (a) at the center plane of the supply fans (b) farther from the center of the supply fans.

Dryer type	Synthesis methods	Number of trays	Air flow speed achieved (m/s)		References
			Baseline geometry	Optimized geometry	
Cabinet dryer	CFD	3	0.79–2.72	0.79–1.95	¹⁶
Indirect solar food dryer	Design development and CFD	4	–	0.05–0.25	¹⁷
Tray dryer	Design and CFD	15	1.49–7.36	2.70–6.30	²⁹
Tray dryer	Design and CFD	21	–	0–0.3	¹⁰
Multi-tray cabinet dryer	Design, CFD and SLP based geometry optimization	32	0.3–1.9	0.5–1.6	This study

Table 6. Comparative summary of recent related studies, highlighting similarities across optimization of supply fan positions for enhanced airflow distribution in multi-tray dryers.

Conclusion

This study presented an integrated Computational Fluid Dynamics (CFD) and Sequential Linear Programming (SLP) approach to optimize airflow distribution in a cabinet-type multi-tray dryer. Using orthogonal experimental design and polynomial regression modelling, the influence of supply fan positions on airflow uniformity was systematically evaluated. The regression model exhibited high predictive accuracy with a coefficient of determination ($R^2 = 0.9247$) and RMSE of 0.0037, validating the robustness of the model. The optimization processes achieved a notable enhancement in the airflow uniformity index (U_i), reaching a maximum value of 76.6% at optimal fan positions. This study achieved a uniform airflow range of 0.82–1.9 m/s, confirming the effectiveness of CFD-based optimization in enhancing drying uniformity. These findings underscore the significance of fans positioning in enhancing the airflow uniformity of multi-tray cabinet dryers. When compared to recent studies, a significant improvement in airflow uniformity has achieved.

To improve the optimization’s convergence behavior, trust-region strategies were introduced. Compared to the SLP method without trust regions, which required approximately 661 iterations, the use of static and dynamic trust-region mechanisms significantly accelerated convergence, reducing the iteration count to 15 and 8, respectively without substantially altering the optimal result. Notably, the dynamic trust-region method demonstrated the fastest convergence rate with minimal zigzagging, offering more efficient navigation of the solution space.

The trust-region approach dynamically controls the step size, eliminating the need for manually enforced small steps and contributing to improved stability and speed. This integrated CFD-SLP-trust region framework provides a scalable and computationally efficient tool for optimizing airflow in drying systems. Its versatility also opens potential for application in other thermal or fluid process optimization tasks across food, pharmaceutical, and chemical industries.

Overall, these findings underscore the critical influence of supply fan positioning on airflow uniformity in multi-tray cabinet dryers. While the CFD-SLP approach provides valuable insights applicable to full-scale operations, the absence of experimental validation and thermal coupling remains a key limitation of this study. The optimum airflow uniformity represents a significant improvement in flow distribution. However, it does not necessarily ensure uniform drying, as temperature uniformity may become the next limiting factor. Therefore, future research should integrate thermal uniformity and perform comprehensive experimental validation under practical conditions to establish a complete and reliable optimization framework for multi-tray drying systems.

Data availability

Data will be made available on request.

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D.A.A.: Writing—review & editing, Writing—original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. F.B.T.: Writing—review & editing, Writing—original draft, Validation, Software, Methodology, Investigation, Formal analysis. Y.D.M. : Writing—review & editing, Methodology, Formal analysis, Software.

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Declarations

Competing interests

The authors declare no competing interests.

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